Lecture 04 Assignments

1) Create the circuit schematic of a **4-line Decoder** using its truth table. Show the **address ranges** (assume that the system has a 16-bit address bus) used by the decoder to select four memory chips (each with size 8K X 8).

2) Create the circuit schematic of an **adder** using its truth table (show both half and full adders). Show a 4-bit adder/subtractor circuit from the full adders (show how it can perform a 4-bit addition and subtraction with the help of 4-bit numbers).

3) Describe how the adder/subtractor (from **question 2**) can be a 4-bit **magnitude comparator**.

4) Show the circuit schematic of a 2-bit **multiplier** and a 2-bit multiplication on the multiplier with the help of 2-bit numbers.

5) [4 Points] Answer the following based on the circuit schematic, which is shown in **Figure1** (where ***A*** and ***B*** are two *N*-bit binary inputs and C_{out} is the carryout bit):

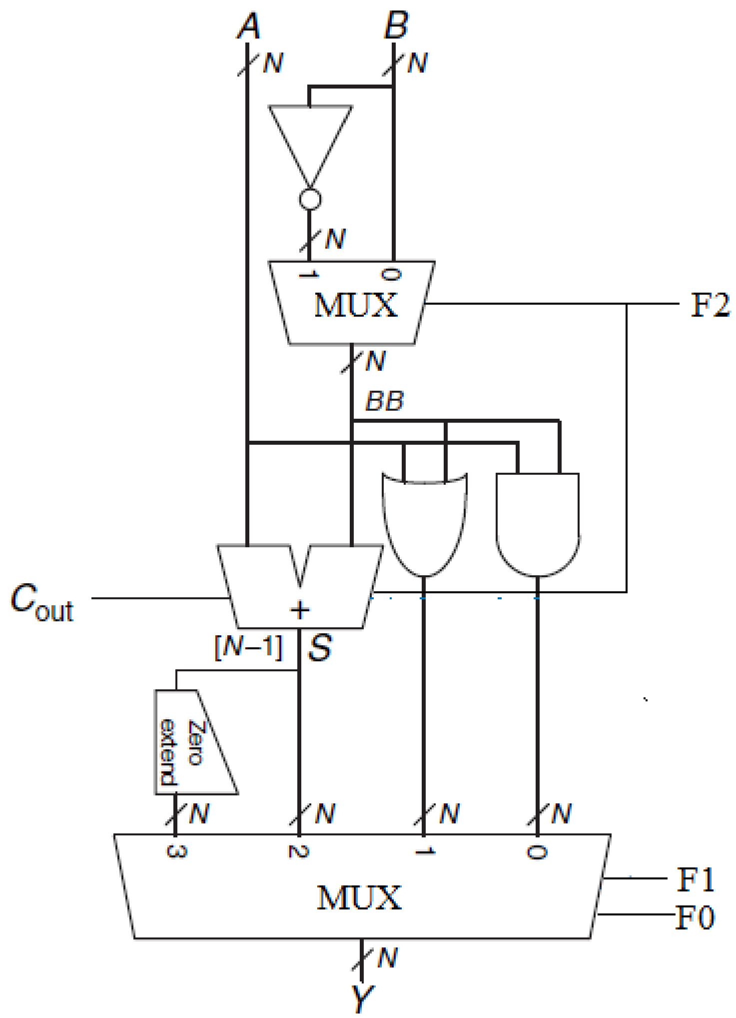

Figure 1.

5.1 (1 point) Identify the circuit.

5.2 (1 point) Find the output, Y when $F_2 = 0$, $F_1 = 1$, and $F_0 = 0$.

5.3 (1 point) Find the output, Y when $F_2 = 1$, $F_1 = 0$, and $F_0 = 1$.

5.4 (1 point) Find the output, Y when $F_2 = 1$, $F_1 = 1$, and $F_0 = 0$.

6) Sketch the schematics of an ***N*-bit ALU** (**Arithmetic and Logic Unit**) with four *N*-bit functions such as *addition*, *subtraction*, *OR operation,* and *parity checker*.

7) Describe any **three main differences** between **SRAM** (Static Random Access Memory) and **DRAM** (Dynamic Random Access Memory). Show a **1-bit DRAM** cell and explain its *read-and-write* operation.

8) Assume that a computer system has a 4Gbyte main memory (organization: 4G x 8). Show the maximum number of address bits used by the system to access the main memory.

9) Briefly describe the following terms:

9.1) Read Only Memory (ROM)

9.2) Erasable Read Only Memory (EROM)

9.3) EEPROM (Electrically Erasable Programmable ROM)

9.3) Flash memory